Physical Chemistry (II)

Lecture 18

Electronic Structure Calculation

Treating Helium

- Perturbation and variational approaches
  - can give very accurate energies
  - computationally difficult
  - not transferable to large atoms and molecules

- Orbital approximation
  - Each electron's spatial part is represented by an one-electron orbital $\phi_i(\vec{r})$
  - Common concept in chemistry
  - How to obtain $\phi_i(\vec{r})$?

Last class

Hartree-Fock Method

- The self-consistent field (SCF) method
- The atomic wavefunction = Slater determinant

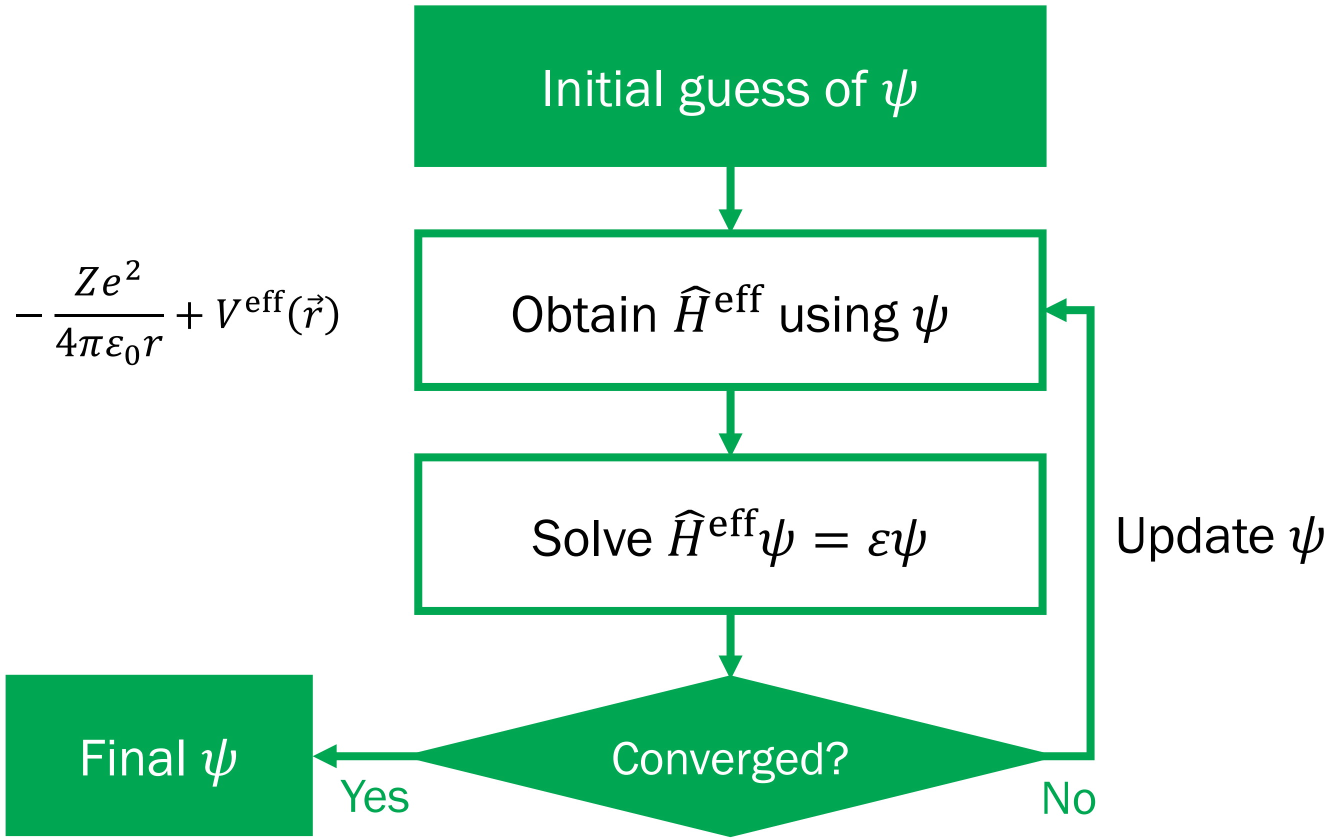
For N quantum states $a, b, \cdots, n$,

$$\phi_{ab\cdots n}(1,2,\cdots,N) = \frac{1}{\sqrt{N!}}\begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

Last class

SCF Method

$$\hat{H}^{\text{eff}} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r} + V^{\text{eff}}(\vec{r})$$



Ground-State Energy

$$E = \sum_{j=1}^{M} 2E_j + \sum_{i=1}^{M} \sum_{j=1}^{M} (2J_{ij} - K_{ij})$$

$$\varepsilon_i = E_i + \sum_{j=1}^{M} (2J_{ij} - K_{ij})$$

$$\therefore \ E = \sum_{i=1}^{M} (E_i + \varepsilon_i)$$

Last class

Koopman's Approximation

$$\text{IE} \approx -\varepsilon_i$$

- Good approximation for closed-shell systems
- Ionization energy gives information about orbital energies
- Photoelectron spectroscopy: experimental method to measure ionization energy

Functional Form of Orbitals

- N-electron Hamiltonian: depends only on distance

$$\hat{H} = \sum_{j=1}^{N}\left(-\frac{\hbar^2}{2m_e}\nabla_j^2 - \frac{Ze^2}{4\pi\varepsilon_0 r_j}\right) + \sum_{j=1}^{N}\sum_{i<j}\frac{e^2}{4\pi\varepsilon_0\left|\vec{r}_i - \vec{r}_j\right|}$$

- Angular part of orbitals: H-atom wavefunctions
  - Spherical harmonics $Y_l^{m_l}(\theta,\phi)$
  - Realized form $\frac{i^{(1\mp1)/2}}{\sqrt{2}}\left\{Y_l^{m_l}(\theta,\phi) \pm \left[Y_l^{m_l}(\theta,\phi)\right]^*\right\}$

Radial Part

- H-atom radial wavefunctions

$$R_{10}(r) = 2\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/a_0}$$

$$R_{20}(r) = \left(\frac{Z}{2a_0}\right)^{3/2} \left(2 - \frac{Zr}{a_0}\right) e^{-Zr/2a_0}$$

$$R_{21}(r) = \frac{1}{\sqrt{3}}\left(\frac{Z}{2a_0}\right)^{3/2} \frac{Zr}{a_0} e^{-Zr/2a_0}$$

- We can use …

$$R_n(r) \propto \left(\frac{\zeta r}{a_0}\right)^{n-1} e^{-\zeta r/a_0}$$

Slater-Type Orbitals (STOs)

$$S_{nlm_l}(r,\theta,\phi;\zeta) = \frac{(2\zeta/a_0)^{3/2}}{\sqrt{(2n)!}}\left(\frac{2\zeta r}{a_0}\right)^{n-1} e^{-\zeta r/a_0} Y_l^{m_l}(\theta,\phi)$$

- ζ: orbital exponent

- Linear combination of STOs can be used to model orbitals
  - e.g. double-zeta orbital

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\vec{r};\zeta_1) + c_2 S_{1s}(\vec{r};\zeta_2)$$

Example: He 1s

For a helium atom, one can use five STOs:

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\zeta_1) + c_2 S_{1s}(\zeta_2) + c_3 S_{1s}(\zeta_3) + c_4 S_{1s}(\zeta_4) + c_5 S_{1s}(\zeta_5)$$

By the Hartree-Fock method, one can optimize

$$\begin{aligned}\phi_{1s}(\vec{r}) = {} & 0.768\, S_{1s}(\zeta = 1.417) + 0.223\, S_{1s}(\zeta = 2.377) \\ & + 0.00408\, S_{1s}(\zeta = 4.396) - 0.00994\, S_{1s}(\zeta = 0.527) \\ & + 0.00230\, S_{1s}(\zeta = 7.843)\end{aligned}$$

Example: He 1s

$$\phi_{1s}(\vec{r}) = 0.768\, S_{1s}(\zeta = 1.417) + 0.223\, S_{1s}(\zeta = 2.377)$$
$$+ 0.00408\, S_{1s}(\zeta = 4.396) - 0.00994\, S_{1s}(\zeta = 0.527)$$
$$+ 0.00230\, S_{1s}(\zeta = 7.843)$$

The optimized energy is

$$E = -2.861680\, E_{\mathrm{h}} \text{ (exp. } -2.9033\, E_{\mathrm{h}})$$

- Determination of ζ takes most of computational time

→ We can "preset" the values of ζ and optimize

M 12.2

Preset Orbital Exponents

Atom	ζ_{1s}	$\zeta_{2s} = \zeta_{2p}$
H	1.24	
He	1.69	
Li	2.69	0.80
Be	3.68	1.15
B	4.68	1.50
C	5.67	1.72
N	6.67	1.95
O	7.66	2.25
F	8.56	2.55
Ne	9.64	2.88

Basis Set

- Atomic orbitals ψ can be represented by linear combinations of some functions ϕ_j

$$\psi(\vec{r}) = \sum_{j=1}^{K} c_j \phi_j = c_1\phi_1 + c_2\phi_2 + \cdots + c_K\phi_K$$

- Basis functions: ϕ_j (e.g. STOs)
- Basis set: $\{\phi_j\}$

Gaussian-Type Orbitals (GTOs)

- In the HF calculation, we must do numerical integration:

$$J_{ij} = \int \psi_i^*(\vec{p})\psi_j^*(\vec{q}) \frac{e^2}{4\pi\varepsilon_0|\vec{p}-\vec{q}|} \psi_i(\vec{p})\psi_j(\vec{q})\, \mathrm{d}\vec{p}\mathrm{d}\vec{q}$$

$$K_{ij} = \int \psi_i^*(\vec{p})\psi_j^*(\vec{q}) \frac{e^2}{4\pi\varepsilon_0|\vec{p}-\vec{q}|} \psi_i(\vec{q})\psi_j(\vec{p})\, \mathrm{d}\vec{p}\mathrm{d}\vec{q}$$

- Gaussian-type orbitals (GTOs): computationally efficient

$$G_{ijk}(r,\theta,\phi;\alpha) = N r^{n-1} e^{-\alpha r^2} Y_l^m(\theta,\phi)$$

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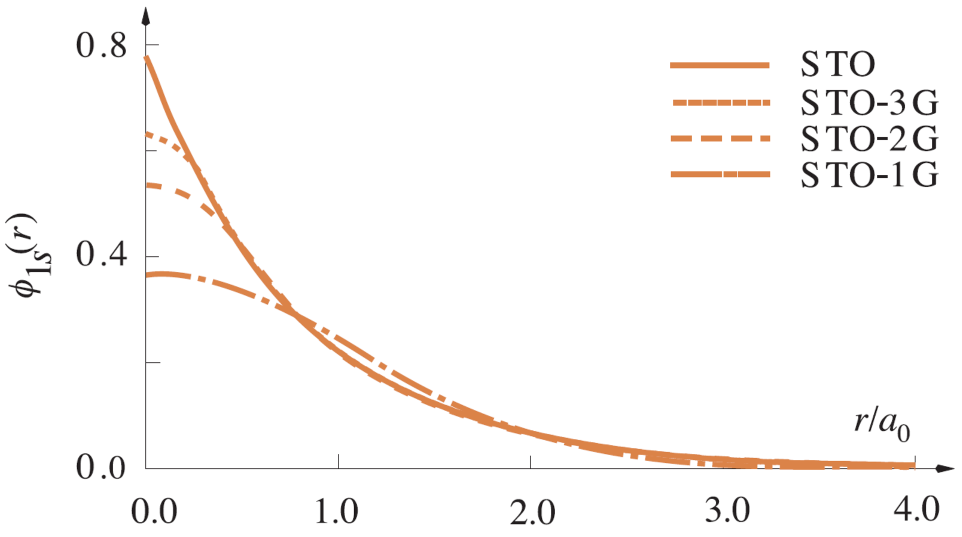
STO-NG: Fitting Trick

$$S_{nlm}(r,\theta,\phi) \leftarrow \sum_{i=1}^{N} d_i G_{nlm}(r,\theta,\phi;\alpha_i)$$



(Image: McQuarrie, ***Quantum Chemistry***, 2nd Ed., Fig. 12.3)

Split-Valence Basis Set

- Inner-shell electrons: a single orbital
- Valence-shell electrons: a sum of orbitals

e.g. Lithium (d_i and α_i are preset)

$$\phi_{1s}(\vec{r}) = d_1 G_{1s}(\alpha_1) + d_2 G_{1s}(\alpha_2) + d_3 G_{1s}(\alpha_3)$$

$$\phi_{2s}(\vec{r}) = c_1 \phi_{2s,1}(\vec{r}) + c_2 \phi_{2s,2}(\vec{r})$$

$$\phi_{2s,1}(\vec{r}) = d_4 G_{2s}(\alpha_4) + d_5 G_{2s}(\alpha_5)$$

$$\phi_{2s,2}(\vec{r}) = G_{2s}(\alpha_6)$$

N-*MP*G Basis Sets

- *N*: number of Gaussian functions for each inner-shell electron orbital
- *M*, *P*: numbers of Gaussian functions for each valence-shell electron orbital
- 3-21G, 6-31G, 6-311G, …

M 12.4

Polarization and Diffuse Functions

- Polarization functions
  - *: *d*-type Gaussian functions added to Li-Ca 2*p* orbitals
  - **: *p*-type Gaussian functions added to H and He 1*s* orbitals
  - 6-31G*, 6-31G**, ...

- Diffuse functions
  - +: Gaussian functions with small exponents on non-H atoms
  - ++: Gaussian functions with small exponents on all atoms
  - 3-21+G, 6-31+G*, 3-21++G, 6-31++G*, ...

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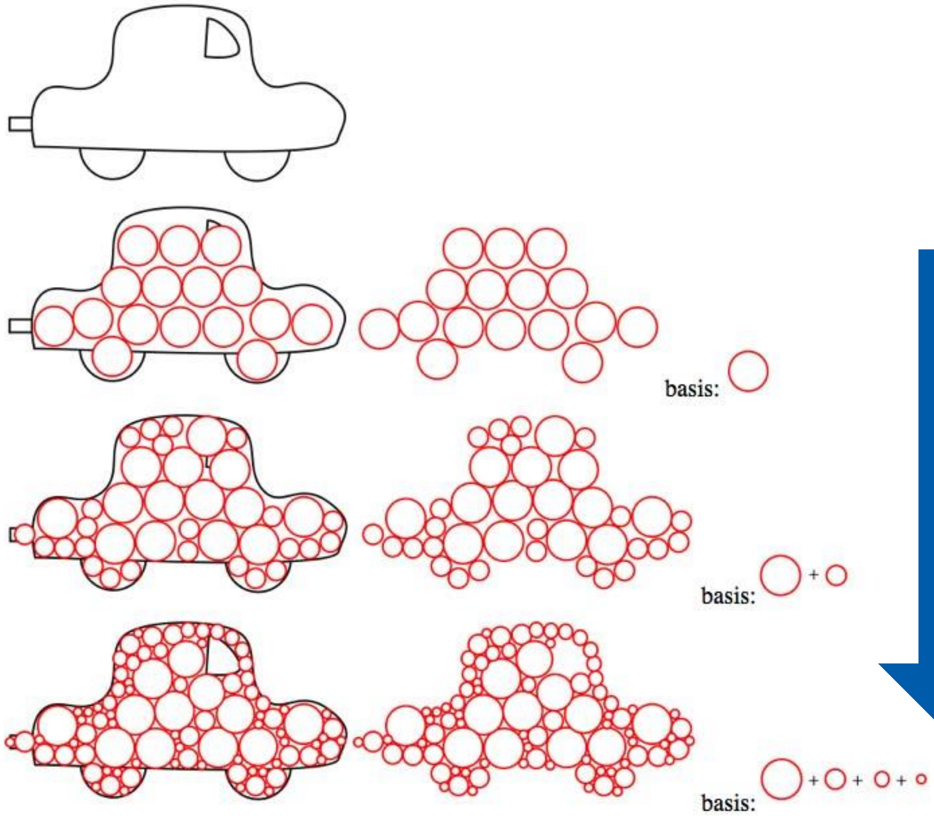
Size of Basis Set



Number of parameters ↑

Accuracy ↑

Computational cost ↑

(Image: http://www.huntresearchgroup.org.uk/teaching/teaching_comp_lab_year2a/4a_pseudo_potentials.html)

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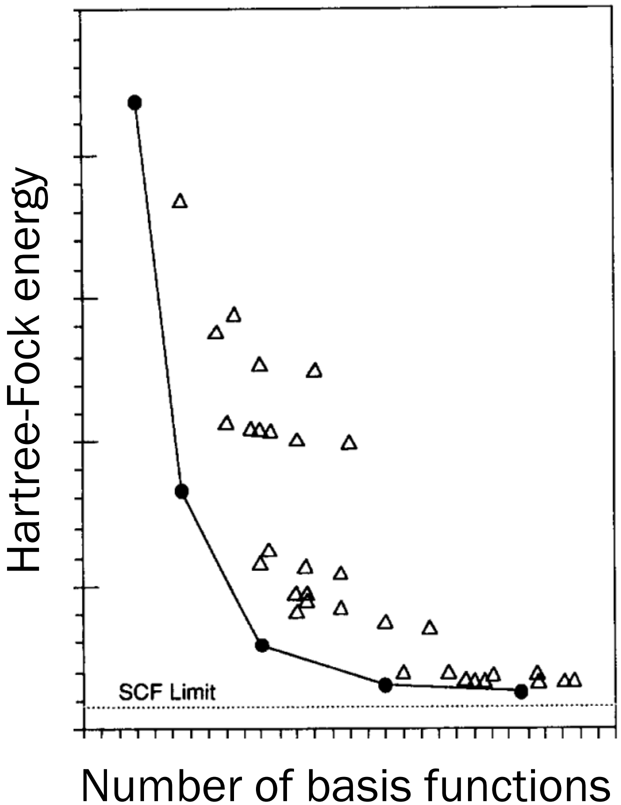
Complete Basis Set (CBS) Limit



(Image: Petersson *et al.*, ***J. Chem. Phys.***, 1988)

RS 9.4
M 9.6

Hartree-Fock Method

- The self-consistent field (SCF) method
- The atomic wavefunction = Slater determinant

- Electrons interact only through an average potential

  $\longrightarrow$ They are "uncorrelated"

- Correlation energy E_{corr}

$$E_{\mathrm{corr}} = E_{\mathrm{exact}} - E_{\mathrm{HF}}$$

Extension of HF

- Start from one-electron wavefunctions ("orbitals") u_i

$$\phi = |u_a \quad u_b \quad \cdots \quad u_n|$$

- HF determinants use occupied orbitals only

- Inclusion of unoccupied (virtual) orbitals may increase accuracy and take care of correlation

Example: Virtual Orbital of He

- Double zeta orbital

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\vec{r}; \zeta = 1.454) + c_2 S_{1s}(\vec{r}; \zeta = 2.911)$$

- Ground-state orbital (by Hartree-Fock)

$$\psi_0 = 0.844\, S_{1s}(\zeta = 1.454) + 0.181\, S_{1s}(\zeta = 2.911)$$

- Virtual orbital (orthogonal to ψ_0)

$$\psi_\mathrm{v} = 1.624\, S_{1s}(\zeta = 1.454) - 1.821\, S_{1s}(\zeta = 2.911)$$

M 9.8

Example: Virtual Orbital of He

- Hartree-Fock wavefunction $\phi_{\mathrm{HF}} = \psi_0(\vec{r}_1)\psi_0(\vec{r}_2)$
- Better wavefunction

$$\phi_{\mathrm{CI}} = c_1\psi_0(\vec{r}_1)\psi_0(\vec{r}_2) + c_2\psi_{\mathrm{v}}(\vec{r}_1)\psi_{\mathrm{v}}(\vec{r}_2)$$

Using a variational method, one can obtain

$$\phi_{\mathrm{CI}} = 0.999\,\psi_0(\vec{r}_1)\psi_0(\vec{r}_2) - 0.0476\,\psi_{\mathrm{v}}(\vec{r}_1)\psi_{\mathrm{v}}(\vec{r}_2)$$

with optimized energy $E = -2.87541\,E_{\mathrm{h}}$

(HF energy $E = -2.86167\,E_{\mathrm{h}}$)

M 12.7

Post-HF Methods

- Configuration interaction (CI)
- Coupled-cluster (CC) theory
- Møller-Plesset (MP) perturbation theory

ab initio methods

- Density functional theory (DFT)

Nobel prize in Chemistry 1998

Configuration Interaction

- Full configuration interaction (full CI)

$$\phi_{\mathrm{CI}} = c_0\phi_0 + \underbrace{\sum_i \sum_a c_i^a \phi_i^a}_{\text{“singles”}} + \underbrace{\sum_{i,j} \sum_{a,b} c_{ij}^{ab} \phi_{ij}^{ab}}_{\text{“doubles”}} + \cdots$$

$\longrightarrow$ Exact solution (in the limit of an infinite basis set)

- Approximations: CI doubles (CID), CI singles and doubles (CISD), …
  - Note: CI singles do not yield correct physics for ground state

Coupled-Cluster Theory

For an N-electron system, the exact wavefunction is

$$\psi_{\text{exact}} = e^{\hat{T}}\phi_0$$

where $\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots + \hat{T}_N$ and $\hat{T}_i$ is an excitation operator.

- CC doubles (CCD)

$$\phi_{\text{CCD}} = e^{\hat{T}_2}\phi_0 = \phi_0 + \hat{T}_2\phi_0 + \frac{1}{2}\hat{T}_2^2\phi_0 + \cdots$$

- CC singles and doubles (CCSD), ...

M 12.7

Møller-Plesset Perturbation Theory

- Unperturbed wavefunction: Hartree-Fock wavefunction

$$\hat{H}_i^{(0)} = -\frac{\hbar^2}{2m_e}\nabla_i^2 - \frac{Ze^2}{4\pi\varepsilon_0 r_i} + V^{\mathrm{eff}}(\vec{r}_i)$$

- Perturbation: difference

$$\hat{H}_i^{(1)} = \sum_j \frac{e^2}{4\pi\varepsilon_0\left|\vec{r}_i - \vec{r}_j\right|} - V^{\mathrm{eff}}(\vec{r}_i)$$

- Notation: MPn (n: order) ⟶ MP2, MP3, MP4, …

Density Functional Theory

- N-particle wavefunction: $\psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$

  $\longrightarrow 3N$ independent variables

- Electron density

$$\rho(\vec{r}) = N \int \psi^*(\vec{r}, \vec{r}_2, \cdots, \vec{r}_N)\psi(\vec{r}, \vec{r}_2, \cdots, \vec{r}_N)\, \mathrm{d}\vec{r}_2 \cdots \mathrm{d}\vec{r}_N$$

$\longrightarrow$ 3 independent variables

Use electron density instead of wavefunction!

Hohenberg-Kohn Theorems

- System energy can be written as a function of ρ:

$$E = E[\rho]$$

- For any trial electron density ρ,

$$E_0 \leq E[\rho]$$

for true ground-state energy E_0 (variational principle).

Kohn-Sham Equation

- Kohn-Sham orbitals: non-interacting "fictitious" particles

$$\rho(\vec{r}) = \sum_i |\phi_i(\vec{r})|^2$$

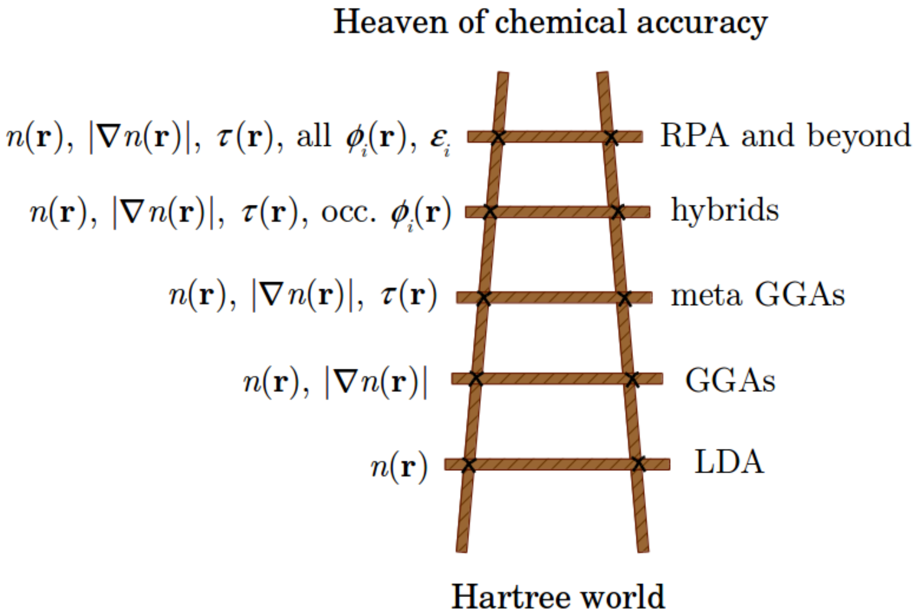
- Kohn-Sham equation

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V^{\text{eff}}\big(\vec{r},\rho(\vec{r})\big)\right]\phi_i(\vec{r}) = \varepsilon_i\phi_i(\vec{r})$$

$$V^{\text{eff}}\big(\vec{r},\rho(\vec{r})\big) = V_{\text{nuc}-e}(\vec{r}) + e^2\int\frac{\rho(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\mathrm{d}\vec{r}\,' + \frac{\delta E^{\text{XC}}[\rho(\vec{r})]}{\delta\rho(\vec{r})}$$

Exchange-Correlation Term

- The exact form of $E^{\mathrm{XC}}[\rho(\vec{r})]$ is unknown
- Many forms have been suggested (LDA, GGA, hybrid, …)



(Image: https://openscholar.huji.ac.il/elikraisler/density-functional-theory)

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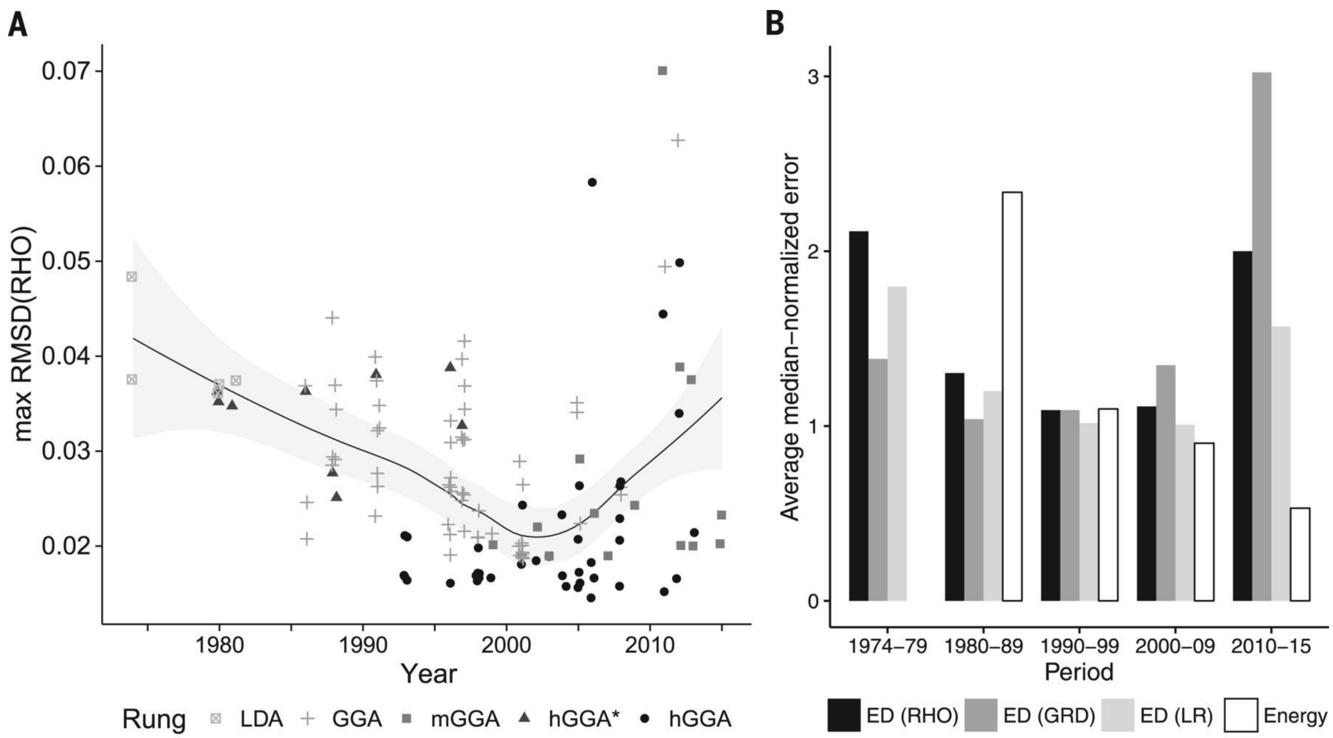
DFT: Are We Doing it All Right?



(Image: Medvedev *et al.*, ***Science*** 2017)

Now We Are Ready…

for Hands-On Practice!